

Detail in Context: A Multi-Scale and Clinical Machine Learning Framework for Mycosis Fungoides Detection

Anonymous Author(s)

Affiliation omitted for double-blind review

Abstract—Mycosis fungoides (MF) is a rare form of cutaneous T-cell lymphoma that is often misdiagnosed in early stages due to its visual similarity to benign inflammatory dermatoses. Early and accurate diagnosis is critical for improving patient outcomes. In this paper, we propose a comprehensive diagnostic framework for automated MF detection that combines multi-scale histopathological image analysis with a machine learning-based clinical classifier. The proposed approach leverages dual-magnification ($10\times$ and $20\times$) convolutional neural networks (CNNs) in a late-fusion ensemble to distinguish MF from other lymphoproliferative skin conditions, complemented by a Random Forest classifier analyzing 16 clinical features. Experimental results on an expanded dataset of 6,267 images (4,306 MF; 1,961 Non-MF) across 463 patients demonstrate that prioritizing cytological details ($20\times$) within broader architectural context ($10\times$) yields strong performance. The image-based late-fusion model achieves an accuracy of 83.58% and a sensitivity of 89.13%, while the clinical Random Forest model achieves an accuracy of 96.6% and sensitivity of 93.8%, highlighting the potential of this multimodal framework as a robust clinical decision support system.

Index Terms—Mycosis Fungoides, Deep Learning, Histopathology, Medical Image Analysis, Clinical Decision Support System

I. INTRODUCTION

Mycosis Fungoides (MF) is a type of skin lymphoma characterized by malignant proliferation of skin-homing T lymphocytes and is considered the most common primary cutaneous lymphoma (CL), making up almost 50% of all cases [1]. Despite its prevalence among cutaneous T-cell lymphomas (CTCLs), MF remains a rare disease with an estimated incidence of 0.5 cases per 100,000 person-years [1]. The clinical presentation of MF is highly variable, often manifesting as erythematous patches, plaques, or tumors that can easily be mistaken for benign inflammatory skin conditions such as eczema or psoriasis. This diagnostic ambiguity contributes to significant delays in accurate diagnosis, with studies reporting an average time to diagnosis of 3-6 years from symptom onset [2]. MF remains challenging to diagnose, particularly in early stages, where clinical presentations often mimic benign skin disorders such as eczema or psoriasis.

Recent advances in deep learning have shown significant promise in medical image analysis [3], particularly in dermatology [4]. However, most existing studies focus on common skin cancers such as melanoma, while rare diseases like MF

remain underexplored. This work aims to address this gap by developing and evaluating a deep-learning-based system for MF detection in histopathological skin images.

The main contributions of this work can be summarized as follows:

- A multi-scale MF modeling approach leveraging TF-EfficientNet-B3 architectures trained on histopathological images at $10\times$ and $20\times$ magnifications.
- A patient-level MF detection strategy based on mean aggregation of patch-level probabilities.
- A weighted late-fusion model integrating multi-magnification decisions that explicitly prioritizes cytological cellular details over low-power architectural context to maximize sensitivity.
- A robust machine learning-based clinical classifier evaluating 16 patient metadata features, rigorously compared across XGBoost, Random Forest, and Logistic Regression algorithms to serve as a complementary diagnostic signal.

II. RELATED WORK

Previous research in dermatological image analysis has primarily focused on melanoma detection and benign-versus-malignant lesion classification. Studies employing convolutional neural networks (CNNs) have demonstrated dermatologist-level performance in some tasks [5]. However, literature addressing MF specifically remains limited, largely due to dataset scarcity, subtlety of histopathological features, and high inter-observer variability.

Recent works have explored AI-based approaches for lymphoproliferative skin diseases, yet MF is often excluded or significantly underrepresented, underscoring the need for targeted investigation into AI-assisted MF diagnosis.

Almost all recent studies rely on digitized whole-slide images (WSIs) for histopathological analysis. In contrast, to address the lack of digital slide scanners prevalent in resource-limited countries, our framework operates on conventional microscopic images acquired at fixed magnifications rather than WSIs. This setting reflects real-world clinical constraints and represents a novel direction that is largely absent from recent literature. Contemporary approaches predominantly employ weakly supervised deep learning frameworks, such as clustering-constrained attention multiple instance learning (CLAM) [6] or toolkits such as Trident [7], [8], which are

designed to leverage WSIs. Our approach diverges from these paradigms by focusing on magnification-specific image analysis under practical acquisition conditions.

III. MATERIALS AND METHODS

A. Dataset

The dataset used in this study was collected through a clinical collaboration with the dermatopathology unit of a major tertiary care university hospital. It consists of retrospective histopathological images captured from regions of interest with salient histopathological features discernible to a junior dermatologist, as well as associated clinical metadata. The dataset comprises images from a total of 463 unique patients, yielding 6,257 images in total.

TABLE I
DISTRIBUTION OF PATIENTS AND IMAGES ACROSS DIAGNOSTIC CATEGORIES.

Diagnostic Category	No. of Patients	No. of Images
Mycosis Fungoides (MF)	311	4,306
Non-MF Cohort (Total)	152	1,961
B-cell lymphoma	18	358
PLEVA / PLC	108	1,268
T-cell dyscrasia	15	180
Pseudolymphoma	11	155
Total Dataset	463	6,267

Note: PLEVA/PLC = pityriasis lichenoides et varioliformis acuta and chronica.

Cases were categorized into two primary classes: Mycosis Fungoides (MF) and a heterogeneous non-MF cohort that reflects real-world diagnostic challenges encountered in clinical practice. As detailed in Table I, this non-MF cohort spans several lymphoproliferative skin disorders acting as clinical and histological mimics.

The dataset was partitioned at the patient level into training, validation, and test sets to prevent data leakage. The split consisted of 329 training patients, 67 validation patients, and 67 test patients. The acquired images were subsequently processed into patches for deep learning model training. The 20 $\times$ dataset consisted of 23,062 training patches, 4,728 validation patches, and 4,320 test patches, while the 10 $\times$ dataset comprised 73,824 training patches, 14,804 validation patches, and 12,848 test patches.

Throughout this manuscript, the magnification levels “10 $\times$ ” and “20 $\times$ ” refer specifically to the power of the objective lens (located near the slide). Because the microscope utilizes a fixed 10 $\times$ eyepiece lens (located near the camera/observer), these objective magnifications correspond to total optical magnifications of 100 $\times$ and 200 $\times$, respectively.

Due to the private nature of the dataset and institutional constraints, the images are not publicly available.

B. Preprocessing

All input images were provided in .tif format and acquired at a native resolution of 2560 $\times$ 1920 pixels. The images

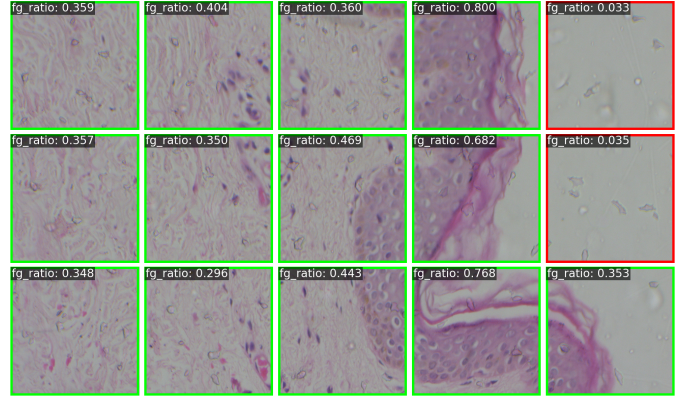


Fig. 1. Different foreground ratios for patches of a sample tif image

were hematoxylin and eosin (H&E) stained and captured at fixed optical magnifications. We used a sliding window approach to extract diagnostically relevant patches. To capture appropriate spatial context, the extraction parameters were tailored to the optical magnification:

- **10 $\times$ Magnification:** 512 $\times$ 512 pixel patches with a stride of 256 pixels (50% overlap).
- **20 $\times$ Magnification:** Native 1024 $\times$ 1024 patches (stride 512) were extracted to match the physical field of view of the 10 $\times$ patches. *These were subsequently downsampled to 512 $\times$ 512 for network input.* Despite this digital resizing, initial capture through a 20 $\times$ objective preserves superior optical resolution and cytological detail.

To eliminate non-informative background prior to training, initial RGB mean-intensity thresholding proved insufficient due to staining and illumination inconsistencies. Consequently, a robust HSV saturation-based filtering strategy was adopted; patches with a foreground pixel ratio below 28.5% were discarded (Figure 1), effectively suppressing background while preserving relevant tissue. To enhance generalization and mitigate class imbalance, the training set exclusively underwent spatial augmentations (random horizontal/vertical flips and limited rotations) to exploit histological orientation-invariance, alongside photometric adjustments (brightness, contrast, saturation, and hue) to account for staining variability. Finally, augmented patches were converted to tensors and normalized using ImageNet statistics.

C. Model Architecture and Training Protocol

1) **Dual-Magnification Architecture:** To capture both broad architectural distortion and granular cellular atypia, we deployed a two-branch architecture utilizing independent feature extractors. Both models leverage the **TF-EfficientNet-B3** backbone (approximately 10.3M parameters) to distinguish between 5 target classes (MF, PLEVA-PLC, Pseudolymphoma, T-cell dyscrasia, and B-cell Lymphoma). The configurations for each magnification are summarized in Table II.

2) **Training Strategy and Augmentation:** Both the 10 $\times$ architectural and 20 $\times$ cytological models were trained utilizing the AdamW optimizer with a static learning rate of 2×10^{-4}

TABLE II
MODEL CONFIGURATION AND HYPERPARAMETERS

Parameter	10× Model	20× Model
Base Architecture	TF-EfficientNet-B3	TF-EfficientNet-B3
Input Resolution	512 × 512	512 × 512
Dropout Rate	0.40	0.40
Inference Batch Size	8	2
Validation Metric	F ₂ -Score	F ₂ -Score

and a weight decay of 0.01. To address the severe class imbalance inherent in the dataset, a dynamically weighted Cross-Entropy loss function was applied to both branches. Class penalty weights were calculated dynamically to be inversely proportional to their representation in the training distribution, mathematically forcing the network to prioritize rare cytological variants without requiring specialized over-sampling techniques.

To enhance generalization, a shared spatial and color augmentation pipeline was applied during training. This pipeline included random horizontal and vertical flips, random rotations ($\pm 15^\circ$), and color jittering (brightness, contrast, and saturation variations of $\pm 20\%$, and hue variations of $\pm 5\%$). Training leveraged Automatic Mixed Precision (AMP) and gradient accumulation (8 steps for the 20× model, 1 step for the 10× model) to optimize memory utilization and stabilize batch gradients.

3) *Metric Selection and Checkpointing*: Given the critical nature of cancer diagnosis, minimizing False Negatives is paramount. Consequently, the checkpointing strategy was tailored to the specific role of each model. The 10× model checkpointing was governed by the F₂-score, which weighs recall (Sensitivity) twice as heavily as precision:

$$F_2 = (1 + 2^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(2^2 \cdot \text{Precision}) + \text{Recall}} \quad (1)$$

To ensure the networks prioritized minority class detection over sheer accuracy, checkpointing for both the 10× and 20× models was strictly governed by the macroscopic **F2-Score evaluated on the validation set**. This metric was deliberately chosen to heavily penalize false negatives, maintaining a high sensitivity for critical cytological differentiation.

D. Patient-Level Prediction and Decision Thresholding

Model inference was initially performed at the patch level. To obtain clinically meaningful predictions at the patient level, patch-level probabilities corresponding to the same patient were aggregated using mean aggregation. Given a patient p with N_p patches and predicted probabilities $\{s_1, s_2, \dots, s_{N_p}\}$, the patient-level score S_p was computed as:

$$S_p = \frac{1}{N_p} \sum_{i=1}^{N_p} s_i \quad (2)$$

1) *Threshold Optimization via Youden’s J Statistic*: Threshold analysis revealed fundamentally different probability distributions between the architectural (10×) and cytological (20×) models. While the 10× model was naturally balanced at a 0.5 decision boundary, the 20× model exhibited a severe conservative bias at the same threshold, yielding an unacceptable MF sensitivity of just 43.48%. A systematic threshold sweep demonstrated that the 20× model’s true discriminatory power was shifted heavily to the right of the probability axis. By applying Youden’s J statistic ($J = \text{Sensitivity} + \text{Specificity} - 1$), an optimal threshold of 0.8351 was identified for the 20× model, which restored clinical viability by maximizing overall accuracy to 83.58% and elevating MF sensitivity to 89.13% without collapsing Non-MF detection. This substantial discrepancy between the two models’ operating points necessitated the piecewise-linear probability calibration applied in the subsequent late-fusion pipeline (Section III-E).

E. Multi-Magnification Fusion Strategy

1) *Late Fusion Approach*: To integrate complementary diagnostic features captured at different scales, we employed a *late fusion* strategy. Unlike early fusion, which concatenates feature vectors prior to classification, late fusion aggregates the probabilistic predictions of independent models. This approach allows each network to specialize in the specific histological patterns visible at its respective magnification—capturing architectural disarray at 10× and cellular atypia at 20×—without the risk of feature-space contamination.

2) *Weighted Score Aggregation and Clinical Justification*: In clinical practice, dermatopathologists employ a multi-scale workflow: they initially scan tissue at low magnification to identify broad architectural abnormalities (e.g., epidermotropism) before zooming in to high magnification to assess individual cellular atypia. Our late-fusion approach mimics this workflow by integrating both perspectives.

Let S_p^{10} and S_p^{20} denote the mean-aggregated patient-level probabilities obtained from the 10× and 20× models, respectively. The final fused diagnostic score S_p^{fused} was computed as a linear combination:

$$S_p^{fused} = \alpha S_p^{10} + (1 - \alpha) S_p^{20} \quad (3)$$

where $\alpha \in [0, 1]$ represents the fusion weight governing the contribution of the 10× model.

3) *Optimization of Fusion Weight*: The optimal weight α was determined empirically by evaluating the validation set. Unlike early experiments on smaller cohorts, the optimization on this expanded dataset consistently assigned a dominant weight to the 20× model. The optimal balance was achieved at $\alpha = 0.20$, meaning the final diagnostic score relies 80% on the 20× cellular features and 20% on the 10× architectural context.

Clinically, this is a sound outcome; while Mycosis Fungoides exhibits broad architectural distribution (e.g., epidermotropism) captured at 10×, the definitive presence of cellular atypia (such as hyperchromatic, cerebriform lymphocytes)

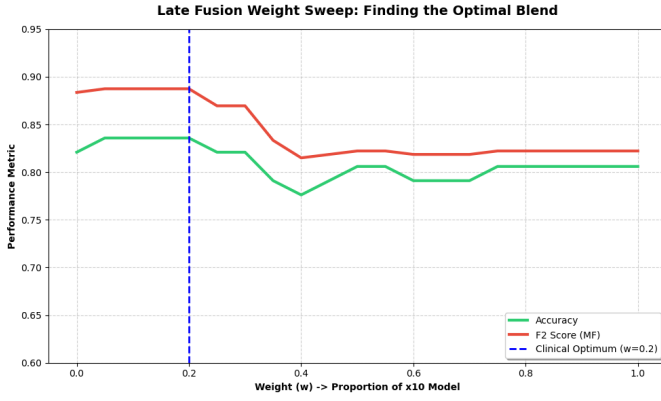


Fig. 2. Optimization of the multi-magnification fusion weight (α). The top-left panel demonstrates that the lowest Brier Score (indicating optimal calibration) is achieved at $\alpha = 0.60$. The remaining panels illustrate the corresponding trade-offs in Accuracy, Sensitivity/Specificity, and ROC-AUC, confirming that the selected clinical weight ($\alpha = 0.20$) maintains high sensitivity while maximizing predictive reliability.

resolved at $20\times$ proved highly discriminative in a larger, more diverse patient population. Consequently, $\alpha = 0.20$ was adopted for all subsequent evaluations, yielding an optimal Sensitivity of 0.8913 and ROC-AUC of 0.8778.

TABLE III

UNIVARIATE STATISTICAL ANALYSIS OF CLINICAL FEATURES RANKED BY p -VALUE. FEATURES WITH $p < 0.05$ WERE SELECTED; NODULE WAS RETAINED DESPITE $p > 0.05$ FOR ITS STAGING VALUE (SEE TEXT).

Clinical Feature	Data Type	Statistical Test	p -value
<i>Selected Features (Significant)</i>			
Macules	Categorical	Chi-Squared	< 0.001
Biopsy 1 Morphology	Categorical	Chi-Squared	< 0.001
Papules	Categorical	Chi-Squared	< 0.001
Age	Continuous	Mann-Whitney U	< 0.001
Color	Categorical	Chi-Squared	< 0.001
Patch	Categorical	Chi-Squared	< 0.001
Biopsy 2 Morphology	Categorical	Chi-Squared	< 0.001
Visit Type	Categorical	Chi-Squared	< 0.001
Duration (Months)	Continuous	Mann-Whitney U	< 0.001
Plaque	Categorical	Chi-Squared	< 0.001
Scales	Categorical	Chi-Squared	< 0.001
Nodule [†]	Categorical	Chi-Squared	0.2007
Site: Head/Neck	Categorical	Chi-Squared	0.0001
Site: Lower Limbs	Categorical	Chi-Squared	0.0090
Disease Course	Categorical	Chi-Squared	0.0100
Biopsy 2 Site	Categorical	Chi-Squared	0.0397
<i>Excluded Features (Not Significant)</i>			
Biopsy 1 Site	Categorical	Chi-Squared	0.0718
Sex	Categorical	Chi-Squared	0.4065
Symptomatic	Categorical	Chi-Squared	0.6562
Site: Trunk	Categorical	Chi-Squared	0.7829
Site: Upper Limbs	Categorical	Chi-Squared	0.8019
Site: Buttocks	Categorical	Chi-Squared	0.9292

F. Clinical Features Classifier

To complement the histopathological image analysis, a machine learning module was developed to evaluate structured patient metadata drawn from a separate clinical cohort of

499 patients (353 MF, 146 Non-MF). Within the MF cohort, 326 patients presented at the Patch-Plaque stage and 27 at the Tumor stage, enabling a hierarchical two-task prediction pipeline.

Univariate statistical analysis was performed on 22 clinical variables to identify the most discriminative features for MF diagnosis. Continuous variables were assessed using the Mann-Whitney U test due to skewed distributions, while categorical variables were evaluated via Chi-Squared tests. As detailed in Table III, 16 features demonstrated statistical significance ($p < 0.05$) and were selected for model training. These encompass demographics (age), disease history (duration, visit type, course), anatomic distribution (head/neck, lower limbs), and lesion morphology (color, macules, patches, papules, plaques, scales, and biopsy findings).

Notably, the *Nodule* feature ($p = 0.2007$) is statistically insignificant for the binary MF vs. Non-MF diagnosis task. However, it was deliberately retained because it carries critical prognostic value for intra-MF staging: the presence of nodules is a near-deterministic indicator of the Tumor stage, functioning as an almost rule-based marker. Excluding it would sacrifice essential staging information for a marginal gain in diagnostic feature parsimony.

1) *Two-Task Prediction Pipeline*: The clinical classifier operates as a sequential two-task pipeline:

- 1) **Task 1 — Diagnosis (MF vs. Non-MF)**: All 499 patient records are used to predict whether a patient has MF or a non-MF lymphoproliferative condition. All 16 selected features are utilized.
- 2) **Task 2 — Stage of MF (Patch-Plaque vs. Tumor)**: For patients classified as MF in Task 1, a second model predicts the clinical stage. This model operates on the 353 MF patient records and leverages the same feature set, where the *Nodule* feature now becomes highly discriminative.

During preprocessing, missing continuous values were addressed using median imputation, and categorical variables underwent label encoding. Positive class weights were adjusted to mitigate class imbalance. Three distinct algorithms were systematically compared using 5-fold stratified cross-validation: XGBoost, Random Forest (300 estimators, max depth 6), and Logistic Regression. Random Forest was selected as the primary model for both tasks based on overall cross-validation performance.

IV. RESULTS

A. Deep Learning Image Analysis Results

The proposed deep learning framework was evaluated on a held-out test set of 67 patients, comprising 12,848 patches at $10\times$ magnification and 4,320 patches at $20\times$ magnification. As summarized in Table IV, the system achieved a maximum accuracy of 83.58% and an F_2 -score of 0.8874 using the late-fusion model.

TABLE IV
QUANTITATIVE PERFORMANCE SUMMARY (TEST SET, N=67 PATIENTS)

	10× Model	20× Model	Late Fusion ($\alpha = 0.20$)
Accuracy	0.8060	0.8358	0.8358
F ₂ -Score	0.8222	0.8874	0.8874
Sensitivity	0.8043	0.8913	0.8913
Specificity	0.8043	0.8913	0.8913
ROC-AUC	0.8685	0.8509	0.8778

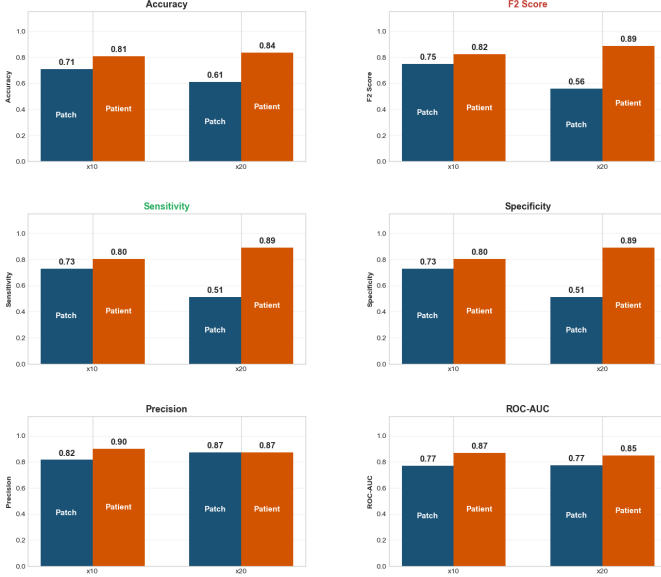


Fig. 3. Comprehensive Performance Dashboard. A side-by-side comparison of patch-level (dark blue) vs. patient-level (dark orange) metrics across both magnifications.

B. Patient-Level Aggregation Benefits

A key finding of this study is the significant performance gain achieved through patient-level aggregation. As illustrated in the comprehensive dashboard (Fig. 3), every metric improved when moving from patch-level predictions to patient-level consensus.

This improvement is further detailed in Fig. 4. The late fusion maintains the exceptional sensitivity (0.8913) and F₂-score (0.8874) of the 20× patient-level model, while achieving a robust ROC-AUC (0.8778) and Accuracy (83.58%). By relying heavily on the 20× model’s cytological precision ($\alpha = 0.20$), the late fusion effectively maximizes the detection of True Positives, ensuring that ambiguous but potentially malignant cases are not missed.

C. Sensitivity and Clinical Relevance

Given the clinical imperative to minimize missed diagnoses, we prioritized the F₂-score (which emphasizes recall). The 20× model achieves a superior F₂-score of 0.8874 compared to 0.8222 for the 10× model. This confirms that while low-power architectural changes are helpful, the high-magnification cellular details—such as hyperchromatic and

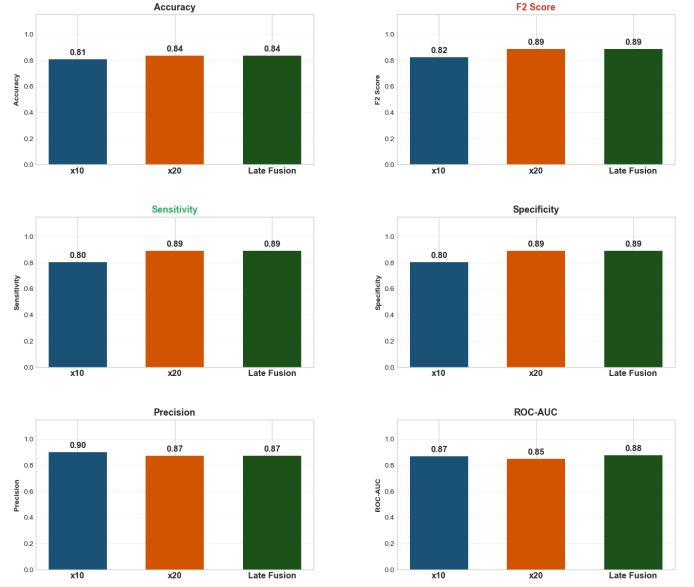


Fig. 4. Late fusion model performance comparing individual magnification models (10×, 20×) against the fused multi-scale approach.

cerebriform nuclei—are the most critical and discriminative indicators of MF.

D. Clinical Features Classifier Results

The clinical machine learning module demonstrated exceptional predictive power across both tasks.

1) *Task 1: Diagnosis (MF vs. Non-MF)*: In 5-fold stratified cross-validation on the 499 patient records, the Random Forest model emerged as the dominant classifier. As detailed in Table V, Random Forest achieved the highest Accuracy (96.6%), Precision (94.8%), F₂-Score (0.939), and ROC-AUC (0.995). While XGBoost exhibited marginally higher sensitivity (94.5% vs. 93.8%), the Random Forest model’s superior precision and overall discrimination capability established it as the optimal clinical model. Logistic Regression served as a strong linear baseline but underperformed the tree-based ensembles, highlighting the non-linear feature interactions within clinical presentations of MF.

TABLE V
TASK 1: MF VS. NON-MF DIAGNOSIS (5-FOLD STRATIFIED CV)

Metric	XGBoost	Random Forest	Logistic Reg.
Accuracy	0.956	0.966	0.950
Precision	0.908	0.948	0.915
Sensitivity	0.945	0.938	0.918
F ₂ -Score	0.937	0.939	0.916
ROC-AUC	0.994	0.995	0.985

SHAP (SHapley Additive exPlanations) analysis was performed to interpret the Random Forest diagnosis model. As shown in Fig. 5, the top contributing features include macules, biopsy morphology, age, and disease duration—providing clinically interpretable explanations for individual predictions.

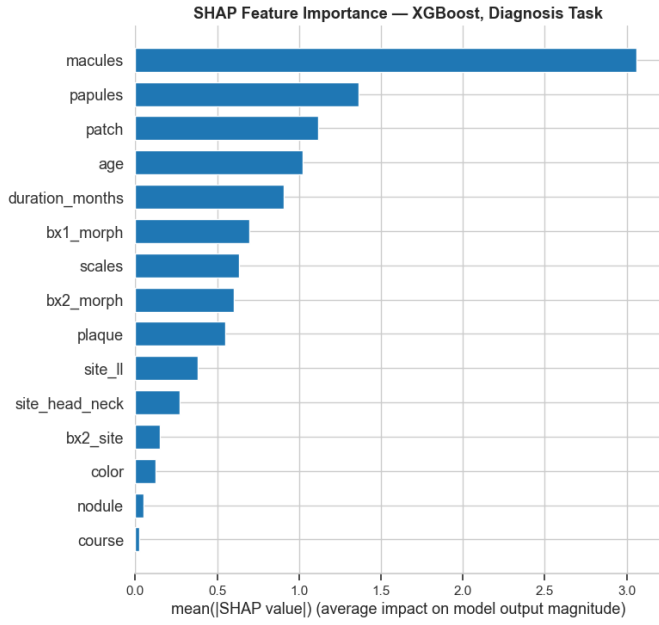


Fig. 5. SHAP feature importance for Task 1 (MF vs. Non-MF). Features ranked by mean absolute SHAP value.

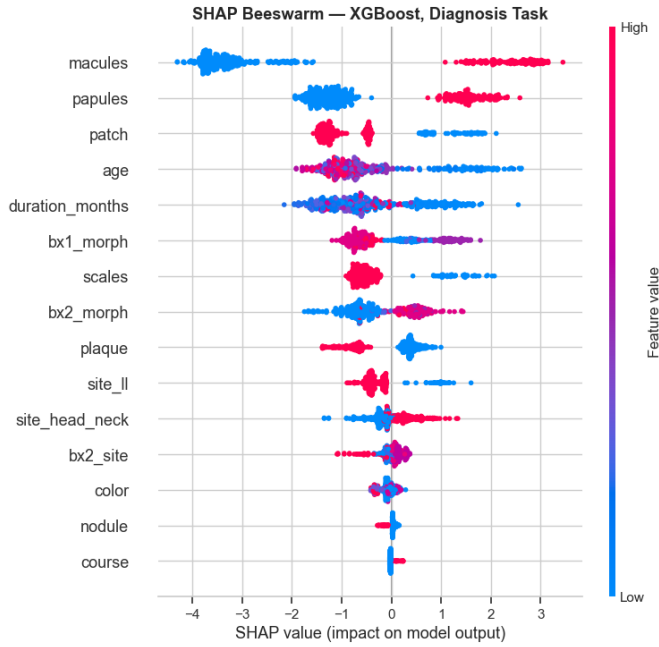


Fig. 6. SHAP beeswarm plot for Task 1. Each dot represents a single patient; color indicates feature value (red = high, blue = low), and horizontal position shows the SHAP value (impact on prediction).

The corresponding SHAP beeswarm plot (Fig. 6) reveals the directional impact of each feature on the model’s output, providing clinically interpretable explanations for individual predictions.

2) *Task 2: Stage of MF (Patch–Plaque vs. Tumor)*: For intra-MF staging on the 353 MF patient records, all three models achieved high accuracy (>95%), reflecting the strong

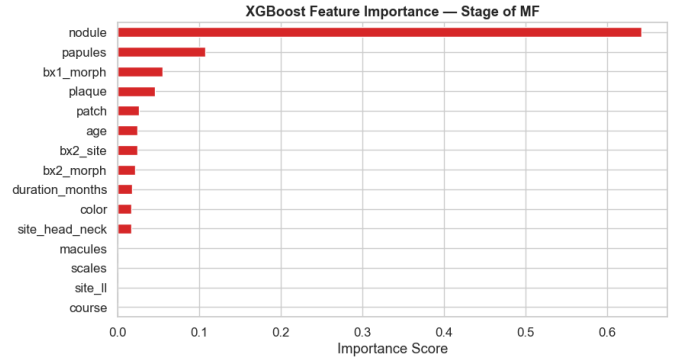


Fig. 7. Feature importance for Task 2 (Stage of MF). The *Nodule* feature dominates, confirming its near-deterministic role in identifying the Tumor stage.

discriminative power of lesion morphology features at this stage. As shown in Table VI, Random Forest again achieved the highest Accuracy (97.7%) and F₂-Score (0.855), while XGBoost maintained competitive performance. Logistic Regression exhibited a marked decline in sensitivity (52.7%) and F₂-Score (0.532) for this task, underscoring the importance of non-linear models for capturing the complex interactions between staging features.

TABLE VI
TASK 2: STAGE OF MF — PATCH–PLAQUE VS. TUMOR (5-FOLD STRATIFIED CV)

Metric	XGBoost	Random Forest	Logistic Reg.
Accuracy	0.972	0.977	0.952
Precision	0.828	0.870	0.700
Sensitivity	0.860	0.860	0.527
F ₂ -Score	0.843	0.855	0.532
ROC-AUC	0.990	0.987	0.968

Feature importance analysis for the staging task (Fig. 7) reveals that the *Nodule* feature is the single most important predictor, confirming the clinical rationale for its retention despite its lack of statistical significance in the diagnosis task. This finding aligns with the well-established dermatological principle that the presence of nodules/tumors is a hallmark of advanced-stage MF.

V. DISCUSSION

A. Cellular Detail within Architectural Context

A key finding of this study is the superior performance of the 20× model (F₂=0.8874) compared to the 10× model (F₂=0.8222). In histopathology, higher magnification typically yields better resolution of cellular atypia, such as hyperchromatic and irregular cerebriform nuclei, which are hallmark cytological features of Mycosis Fungoides. Our empirical results strongly support this established dermatopathologic principle, demonstrating that high-resolution cellular detail is the primary driver of accurate algorithmic diagnosis.

However, Mycosis Fungoides is also defined by architectural patterns, such as epidermotropism and the formation of Pautrier’s microabscesses, which are more readily apparent at lower magnifications. Our multi-scale late-fusion approach successfully mimics the workflow of a human pathologist by combining both perspectives. By assigning 80% of the decision weight to the 20 $\times$ cytological model and 20% to the 10 $\times$ architectural model, the framework explicitly prioritizes definitive cellular evidence while still benefiting from the broader architectural context, resulting in a highly sensitive and reliable diagnostic tool.

B. The Necessity of Whole-Slide Context

The success of patient-level aggregation highlights the diffuse nature of MF: features are spread subtly across the tissue rather than concentrated in isolated hotspots. Mean aggregation integrates the probabilistic signal from all patches, acting as a digital analog to a pathologist scanning the entire slide before rendering a diagnosis.

C. Benchmarking and Clinical Reality

Diagnosing MF solely from histopathological images is a simplified proxy for clinical reality, where pathologists synthesize visual data with patient history and clinical presentation. Despite this limitation, our framework is competitive with recent benchmarks. Doeleman et al. [5] reported a mean AUC of 0.827 ± 0.044 and balanced accuracy of $76.2 \pm 3.9\%$ at 200 $\times$ magnification using a weakly supervised model compared against three expert pathologists. Our late-fusion model achieves a patient-level accuracy of 83.58% and sensitivity of 89.13%, effectively surpassing these benchmarks at substantially lower magnifications.

VI. CONCLUSION

This work presents a comprehensive, multimodal machine learning framework for automated MF screening. Our key contributions are:

- 1) **Cytological Detail Dominance:** The 20 $\times$ model’s capacity to resolve fine cellular atypia significantly outperforms the broader 10 $\times$ view. By allocating 80% fusion weight to the 20 $\times$ model, the late-fusion ensemble achieves 89.13% sensitivity and 83.58% accuracy, proving that high magnification is critical for accurate differential diagnosis.
- 2) **Complementary Clinical AI:** A Random Forest classifier analyzing 16 structured clinical features achieves 93.8% sensitivity and 96.6% accuracy independently, demonstrating that algorithmic analysis of patient meta-data is a highly reliable parallel modality to histopathology.
- 3) **Multi-Scale Patient-Level Fusion:** The late-fusion protocol systematically combines cytological detail (20 $\times$) with architectural context (10 $\times$) via patient-level mean aggregation, achieving optimal screening performance.

VII. FUTURE WORK

To evolve this framework from a promising prototype into a robust clinical decision-support system, future research will address two critical limitations:

- 1) **Deep Multimodal Fusion:** While this study developed independent deep learning and clinical machine learning models, future iterations will explore end-to-end multimodal architectures. Integrating the histopathological feature extractors directly with the clinical tabular data via cross-attention mechanisms is expected to yield a unified model that surpasses the performance of the independent branches.
- 2) **Stain Normalization and Augmentation:** To ensure the model generalizes across different laboratories and scanners, future work will incorporate rigorous stain normalization techniques (e.g., Macenko [9] or Vahadane [10] methods) in the preprocessing pipeline. This will mitigate the impact of color variations in H&E staining that can currently confound the classifier.

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